



OMMAX CASE STUDY

DATA AND AI – Data Science & GenAI

Part A: Classic ML

Business context

The Münchner Verkehrsgesellschaft GmbH hired OMMAX to forecast the number of passengers entering each subway station in Munich hourly (7am–12pm), 7 days in advance, in order to allocate operational resources to the right stations ahead of time.

Dataset

You are given 4 tables:

- station_metadata
- timeseries_with_target
- traffic_hourly
- weather_hourly

Task

Train a model to predict **hourly passenger counts per station, 7 days ahead**.

Train / test split

- Train: all data except the last 7 days
- Test: last 7 days only (**420 rows**)

Evaluation

Report:

- **Overall RMSE** on the test set
- **RMSE per station** on the test set

Production assumptions

- *station_metadata*, *timeseries_with_target* are stored in an Azure Data Lake and updated hourly
- *traffic_hourly* and *weather_hourly* are accessed via APIs and updated hourly
- Accurate weather forecasts are available for the next 10 days

Deliverable

5/10-minute CEO-level slide presentation covering:

- Problem framing



- Feature & model choice
- Evaluation results
- Production deployment & monitoring

Part B: Agentic & GenAI

Business context

A B2B company receives customer orders via unstructured emails that vary in format, language, and quality. Some emails may be incomplete, ambiguous, or malicious. Attachments can include PDF, PPTX, and XLSX files. Today, human operators manually extract order details and enter them into the ERP, making the process slow and hard to scale.

Task

Develop a (1) **notebook** (e.g. jupyter) and a (2) **high-level slide presentation** that demonstrates an example agentic AI workflow capable of converting incoming order emails into ERP orders while ensuring the process remains safe, auditable, and controllable.

Constraints

- All orders arrive in a **single email inbox**
- ERP system exposes a **REST API**
 - Read products, prices, customers, historical purchases
 - Create / update / delete orders

Key points to address

Your notebook should include code that demonstrates:

- End-to-end workflow design (LLM API, orchestration, tools, model selection)
- Data extraction from emails and attachments (PDF, PPTX, XLSX)
- Logging, storage, and auditability (traceability, tool calls, and decision records)
- Clear separation of LLM-based and rule-based responsibilities
- Validation, security controls, and prevention of prompt injection or malicious actions
- Error handling and escalation of incomplete or ambiguous orders

Note: You may use dummy data for emails, documents, and ERP API responses. Data can be stored locally (e.g., database, CSV, JSON, or files).



In addition, include a markdown section at the end covering:

- Deployment architecture and setup (runtime, infrastructure, database, CI/CD etc.)
- Human-in-the-loop workflow design
- Production monitoring and key performance metrics

Deliverable

1. AI Workflow Notebook

2. 15-20 Minute C-Level Slide Presentation

A slide deck explaining:

A. High-Level Solution Design

- Overall architecture
- Agent behavior and decision flow
- Deployment considerations

B. Business Case calculation/estimation

- High-level estimation for C-level about the impact of the solution

Good luck with the preparation!